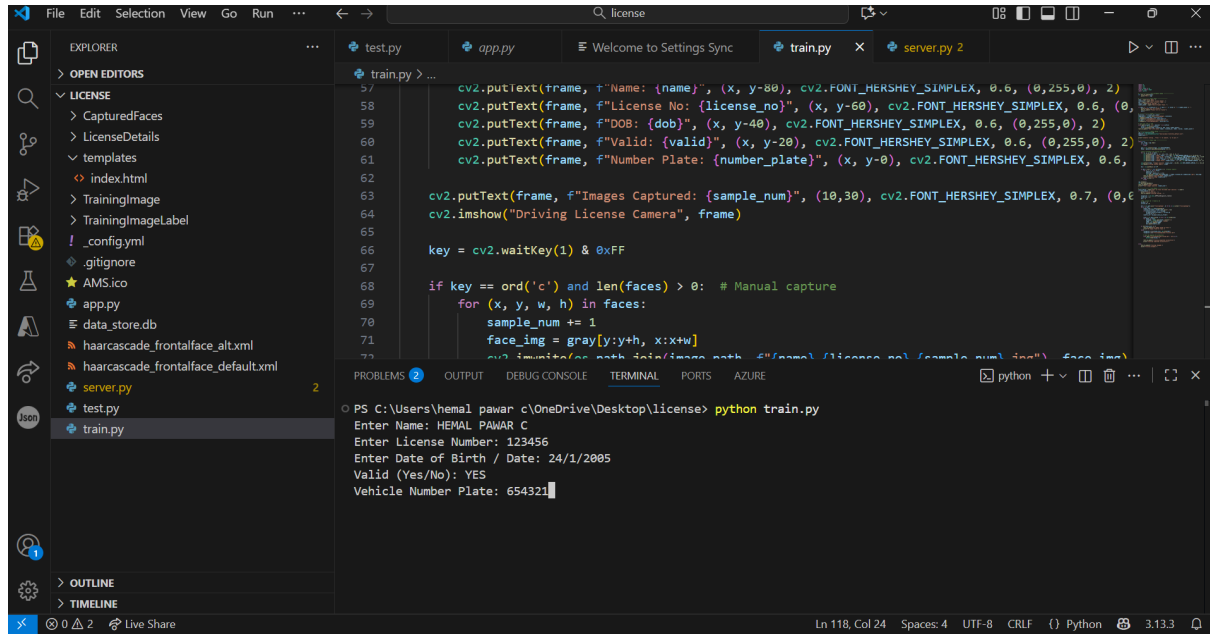


train.py --- FOR TRAINING THE MODEL

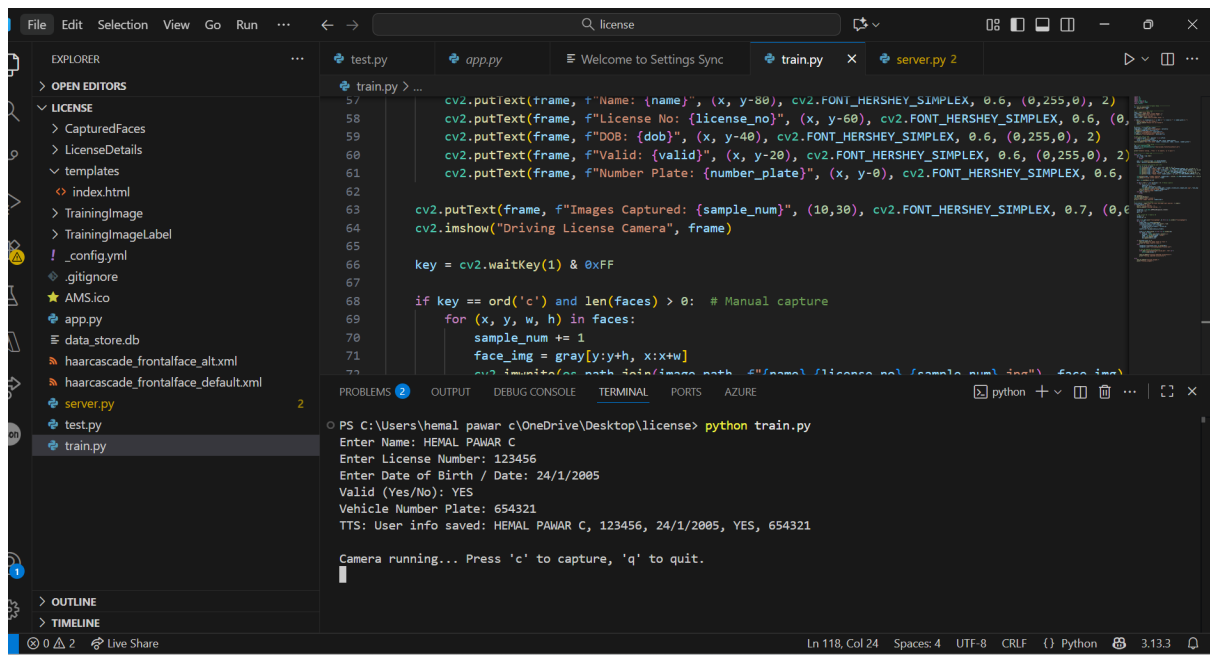


The screenshot shows the Visual Studio Code editor with the `train.py` file open. The file contains Python code for training a model, including functions for capturing faces and saving data. The terminal output shows the execution of `python train.py`, which prompts the user for input and displays the results.

```
57 cv2.putText(frame, f"Name: {name}", (x, y-80), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
58 cv2.putText(frame, f"License No: {license_no}", (x, y-60), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,
59 cv2.putText(frame, f"DOB: {dob}", (x, y-40), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
60 cv2.putText(frame, f"Valid: {valid}", (x, y-20), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
61 cv2.putText(frame, f"Number Plate: {number_plate}", (x, y-0), cv2.FONT_HERSHEY_SIMPLEX, 0.6,
62
63 cv2.putText(frame, f"Images Captured: {sample_num}", (10,30), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,
64 cv2.imshow("Driving License Camera", frame)
65
66 key = cv2.waitKey(1) & 0xFF
67
68 if key == ord('c') and len(faces) > 0: # Manual capture
69     for (x, y, w, h) in faces:
70         sample_num += 1
71         face_img = gray[y:y+h, x:x+w]
72         cv2.imwrite(os.path.join(image_path, f"{name}/{license_no}/{sample_num}.jpg"), face_img)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
PS C:\Users\hemal pawar c\OneDrive\Desktop\license> python train.py
Enter Name: HEMAL PAWAR C
Enter License Number: 123456
Enter Date of Birth / Date: 24/1/2005
Valid (Yes/No): YES
Vehicle Number Plate: 654321
```



The screenshot shows the Visual Studio Code editor with the `train.py` file open. The file contains Python code for training a model, including functions for capturing faces and saving data. The terminal output shows the execution of `python train.py`, which prompts the user for input and displays the results.

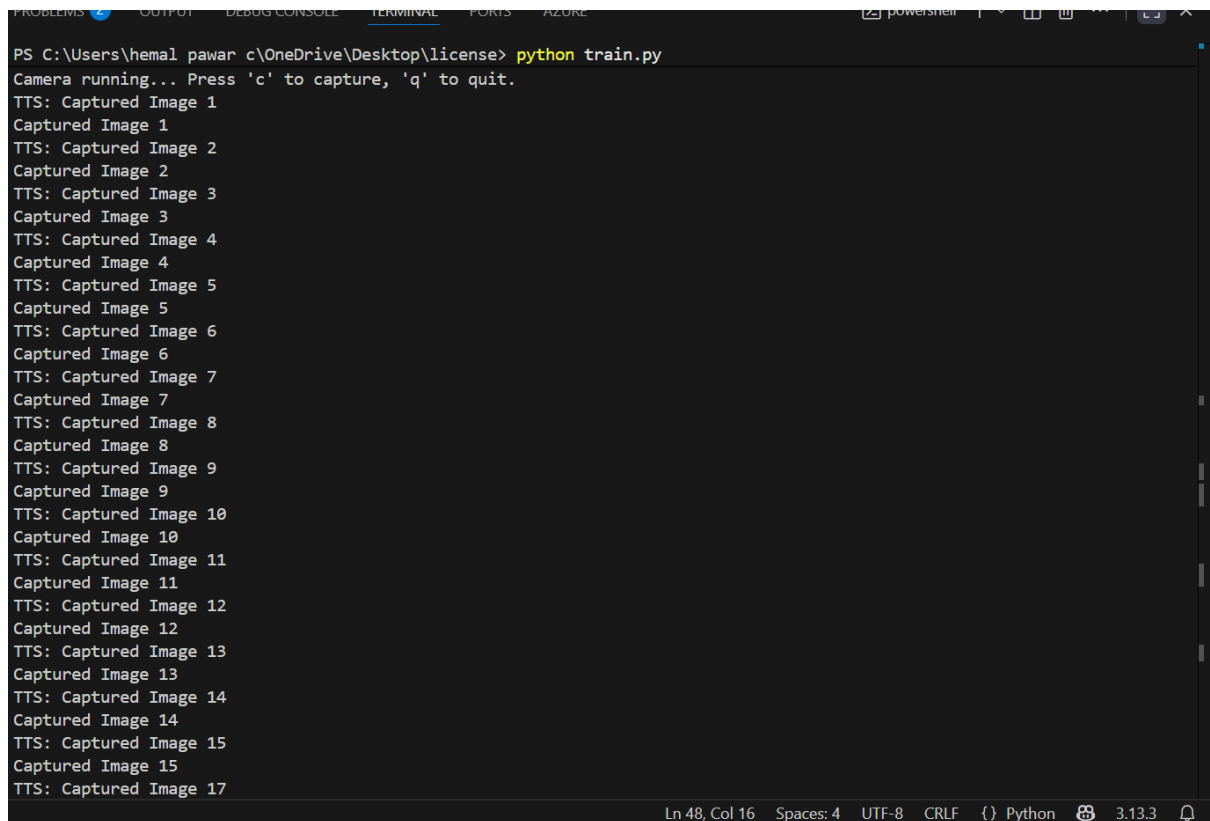
```
57 cv2.putText(frame, f"Name: {name}", (x, y-80), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
58 cv2.putText(frame, f"License No: {license_no}", (x, y-60), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,
59 cv2.putText(frame, f"DOB: {dob}", (x, y-40), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
60 cv2.putText(frame, f"Valid: {valid}", (x, y-20), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
61 cv2.putText(frame, f"Number Plate: {number_plate}", (x, y-0), cv2.FONT_HERSHEY_SIMPLEX, 0.6,
62
63 cv2.putText(frame, f"Images Captured: {sample_num}", (10,30), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,
64 cv2.imshow("Driving License Camera", frame)
65
66 key = cv2.waitKey(1) & 0xFF
67
68 if key == ord('c') and len(faces) > 0: # Manual capture
69     for (x, y, w, h) in faces:
70         sample_num += 1
71         face_img = gray[y:y+h, x:x+w]
72         cv2.imwrite(os.path.join(image_path, f"{name}/{license_no}/{sample_num}.jpg"), face_img)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

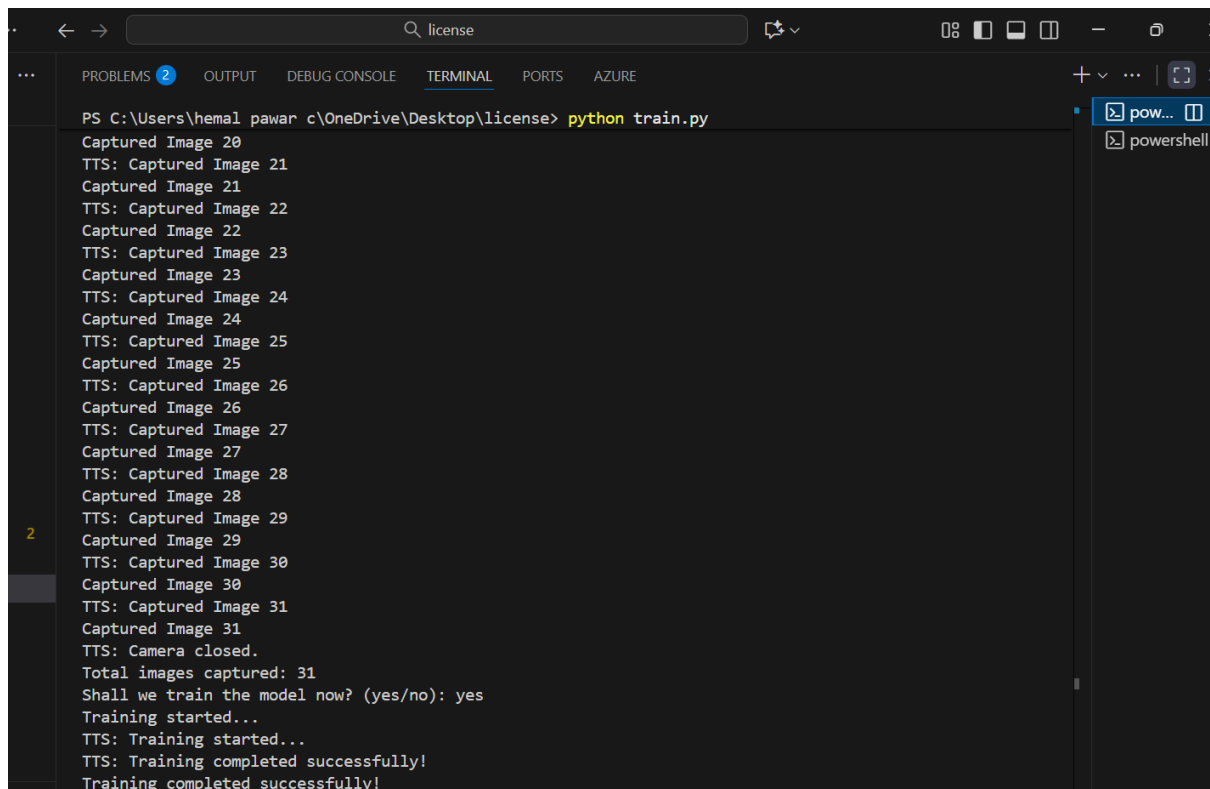
```
PS C:\Users\hemal pawar c\OneDrive\Desktop\license> python train.py
Enter Name: HEMAL PAWAR C
Enter License Number: 123456
Enter Date of Birth / Date: 24/1/2005
Valid (Yes/No): YES
Vehicle Number Plate: 654321
TTS: User info saved: HEMAL PAWAR C, 123456, 24/1/2005, YES, 654321

Camera running... Press 'c' to capture, 'q' to quit.
```

train.py --- FOR TRAINING THE MODEL



```
PS C:\Users\hemal pawar c:\OneDrive\Desktop\license> python train.py
Camera running... Press 'c' to capture, 'q' to quit.
TTS: Captured Image 1
Captured Image 1
TTS: Captured Image 2
Captured Image 2
TTS: Captured Image 3
Captured Image 3
TTS: Captured Image 4
Captured Image 4
TTS: Captured Image 5
Captured Image 5
TTS: Captured Image 6
Captured Image 6
TTS: Captured Image 7
Captured Image 7
TTS: Captured Image 8
Captured Image 8
TTS: Captured Image 9
Captured Image 9
TTS: Captured Image 10
Captured Image 10
TTS: Captured Image 11
Captured Image 11
TTS: Captured Image 12
Captured Image 12
TTS: Captured Image 13
Captured Image 13
TTS: Captured Image 14
Captured Image 14
TTS: Captured Image 15
Captured Image 15
TTS: Captured Image 17
```



```
PS C:\Users\hemal pawar c:\OneDrive\Desktop\license> python train.py
Captured Image 20
TTS: Captured Image 21
Captured Image 21
TTS: Captured Image 22
Captured Image 22
TTS: Captured Image 23
Captured Image 23
TTS: Captured Image 24
Captured Image 24
TTS: Captured Image 25
Captured Image 25
TTS: Captured Image 26
Captured Image 26
TTS: Captured Image 27
Captured Image 27
TTS: Captured Image 28
Captured Image 28
TTS: Captured Image 29
Captured Image 29
TTS: Captured Image 30
Captured Image 30
TTS: Captured Image 31
Captured Image 31
TTS: Camera closed.
Total images captured: 31
Shall we train the model now? (yes/no): yes
Training started...
TTS: Training started...
TTS: Training completed successfully!
Training completed successfully!
```

train.py --- FOR TRAINING THE MODEL

Code:

```
import os

import cv2

import csv

import numpy as np

from PIL import Image

import pickle


# ----- Text-to-Speech Dummy -----

def text_to_speech(msg):

    print("TTS:", msg)


# ----- User Input -----

name = input("Enter Name: ")

license_no = input("Enter License Number: ")

dob = input("Enter Date of Birth / Date: ")

valid = input("Valid (Yes/No): ")

number_plate = input("Vehicle Number Plate: ")


if name == "" or license_no == "" or dob == "" or valid == "" or number_plate == "":

    print("Please fill all fields!")

    text_to_speech("Please fill all fields!")

    exit()


# ----- Paths -----

directory = f"{license_no}_{name}"

image_path = os.path.join("TrainingImage", directory)

os.makedirs(image_path, exist_ok=True)

os.makedirs("LicenseDetails", exist_ok=True)

csv_path = "LicenseDetails/license_details.csv"

os.makedirs("TrainingImageLabel", exist_ok=True)
```

train.py --- FOR TRAINING THE MODEL

```
# Save user info in CSV

with open(csv_path, "a+", newline="") as csvFile:

    writer = csv.writer(csvFile)

    writer.writerow([name, license_no, dob, valid, number_plate])

text_to_speech(f"User info saved: {name}, {license_no}, {dob}, {valid}, {number_plate}")


# ----- Camera Setup -----

cap = cv2.VideoCapture(0)

detector = cv2.CascadeClassifier("haarcascade_frontalface_default.xml")

sample_num = 0


print("\nCamera running... Press 'c' to capture, 'q' to quit.")


# ----- Main Loop -----

while True:

    ret, frame = cap.read()

    if not ret:

        break


    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    faces = detector.detectMultiScale(gray, 1.3, 5)


    # Draw rectangles and overlay user info

    for (x, y, w, h) in faces:

        cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)

        cv2.putText(frame, f"Name: {name}", (x, y-80), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)

        cv2.putText(frame, f"License No: {license_no}", (x, y-60), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)

        cv2.putText(frame, f"DOB: {dob}", (x, y-40), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)

        cv2.putText(frame, f"Valid: {valid}", (x, y-20), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)

        cv2.putText(frame, f"Number Plate: {number_plate}", (x, y-0), cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0,255,0), 2)
```

train.py --- FOR TRAINING THE MODEL

```
cv2.putText(frame, f"Images Captured: {sample_num}", (10,30), cv2.FONT_HERSHEY_SIMPLEX, 0.7,
(0,0,255), 2)

cv2.imshow("Driving License Camera", frame)

key = cv2.waitKey(1) & 0xFF

if key == ord('c') and len(faces) > 0: # Manual capture
    for (x, y, w, h) in faces:
        sample_num += 1
        face_img = gray[y:y+h, x:x+w]
        cv2.imwrite(os.path.join(image_path, f"{name}_{license_no}_{sample_num}.jpg"), face_img)
        text_to_speech(f"Captured Image {sample_num}")
        print(f"Captured Image {sample_num}")
    elif key == ord('q'):
        break

cap.release()

cv2.destroyAllWindows()

text_to_speech("Camera closed.")

print(f"Total images captured: {sample_num}")

# ----- Train Model -----

train_choice = input("Shall we train the model now? (yes/no): ").lower()

if train_choice == "yes":
    print("Training started...")
    text_to_speech("Training started...")

recognizer = cv2.face.LBPHFaceRecognizer_create()

faces_list = []

Ids = []
```

train.py --- FOR TRAINING THE MODEL

```
# Map License No -> Numeric ID

id_map = {}

current_id = 0


dirs = [os.path.join("TrainingImage", d) for d in os.listdir("TrainingImage")]

for d in dirs:

    folder_name = os.path.basename(d)

    license_no_folder = folder_name.split("_")[0]

    if license_no_folder not in id_map:

        id_map[license_no_folder] = current_id

        current_id += 1

    numeric_id = id_map[license_no_folder]


files = [os.path.join(d, f) for f in os.listdir(d)]

for fpath in files:

    pillImage = Image.open(fpath).convert("L")

    imgNp = np.array(pillImage, "uint8")

    faces_list.append(imgNp)

    Ids.append(numeric_id)


if len(faces_list) == 0:

    text_to_speech("No images found to train.")

    print("No images found to train.")

else:

    recognizer.train(faces_list, np.array(Ids))

    recognizer.save("TrainingImageLabel/Trainer.yml")


# Save ID mapping for prediction

with open("TrainingImageLabel/id_map.pkl", "wb") as f:

    pickle.dump(id_map, f)
```

train.py --- FOR TRAINING THE MODEL

```
text_to_speech("Training completed successfully!")
```

```
print("Training completed successfully!")
```

```
else:
```

```
text_to_speech("Training skipped.")
```

```
print("Training skipped.")
```